Final Project: Description Using Models

UCB MIDS 203 Fall 2026 Section 6 Final Project Report

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1. Introduction

Our goal is to describe the relationship between life expectancy and economic development across countries, and to assess how this relationship changes when accounting for demographic and inequality factors. Our audience is individuals without a formal background in socioeconomics who are interested in better understanding how different societal metrics relate to life expectancy and broader patterns in global development and living standards. The larger context of the question lies in discussions of economic development, global inequality, and public health, where researchers and policymakers aim to understand why countries experience different health outcomes and living standards.

Prior research has examined these relationships in depth. Samuel Preston (1975) introduced the Preston curve, which demonstrates a concave relationship between life expectancy and GDP per capita. More recent work, such as Roffia et al., uses longitudinal data to study how socioeconomic and demographic factors influence life expectancy over time.

Our report builds on this literature by combining multiple dimensions of national development into an accessible, country level framework. This approach provides a more holistic view of how economic, demographic, and inequality related factors interact to shape life expectancy.

2. Source Data

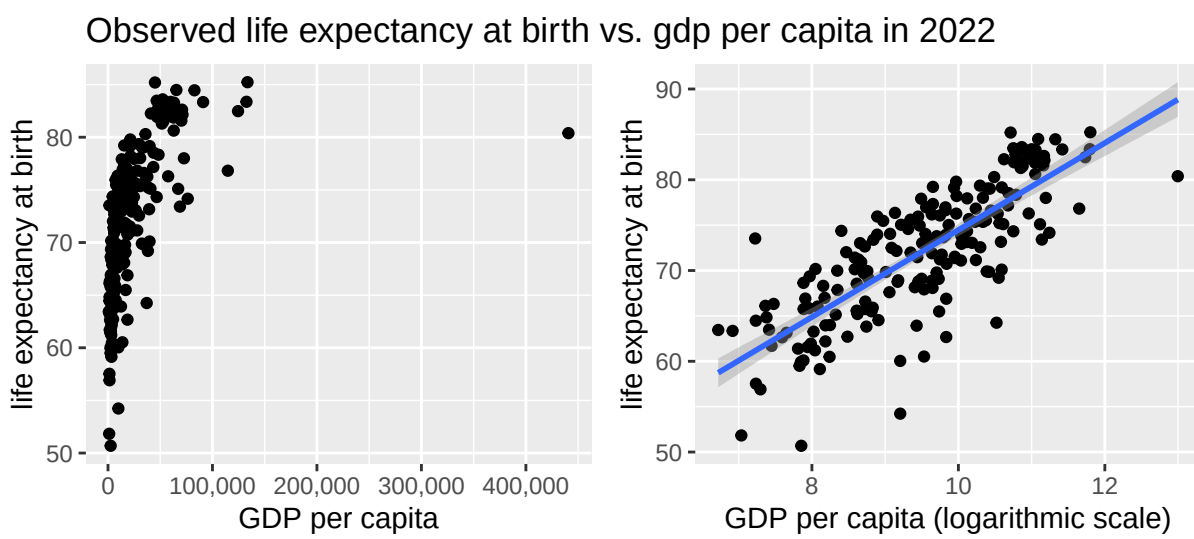
Our data is sourced from Gapminder, a Swedish organization that curates reliable data on global trends, compiled from sources such as the World Bank. The data includes annual country level observations spanning multiple decades. We merge multiple indicator datasets by country identifiers, retain 2022 observations, and construct a cross sectional dataset after removing observations with missing GDP per capita. We use normalized country level metrics (e.g. per capita) to account for differences in population size.

Each country is represented by a three letter country code and a country name. The outcome variable in our analysis is life expectancy (**lex**), which captures overall population health. Our primary explanatory variable is GDP per capita (**gdp_pcap**), which measures economic development. Although the dataset includes hundreds of indicators, we select a subset to represent key dimensions of development, including health, inequality, population demographics, the economy, and infrastructure.:

children_per_woman_total_fertility	average_daily_income
population_density_per_square_k	gini coefficient
cell_phones_per_100_people	median_age_years
broadband_subscribers_per_100_p	co2_pcap_cons

3. Primary Analysis

Our primary analysis examines the relationship between life expectancy at birth and GDP per capita across countries in 2022. GDP per capita is a currency based measure and is typically well suited to a logarithmic transformation. We plot life expectancy against GDP per capita and observe that applying a logarithmic transformation produces a more linear relationship compared to the untransformed data.:



The correlation between life expectancy and the logarithm of GDP per capita is 0.796, supporting the strong linear relationship observed in the scatter plot. We then estimate an Ordinary Least Squares model, regressing life expectancy at birth on GDP per capita. The model is specified as:

$$\text{lex} = 26.501 + 4.796 \cdot \log(\text{GDP per capita})$$

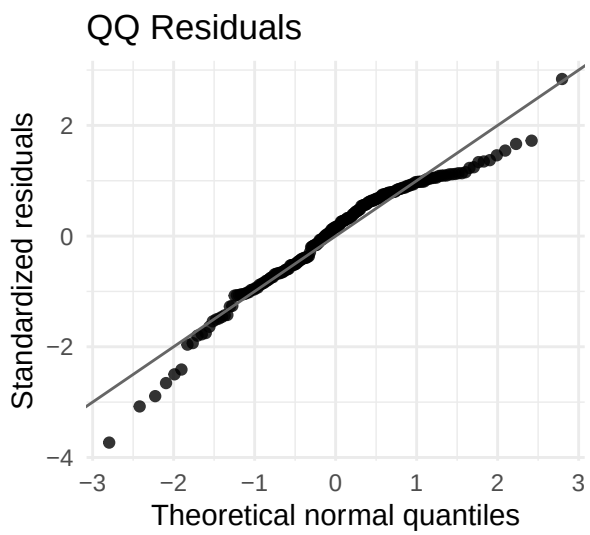
The residuals versus fitted plot shows slight curvature rather than a random scatter around zero, indicating potential misspecification in the functional form. This suggests that GDP per capita alone does not fully capture the relationship and that additional predictors may improve model fit.

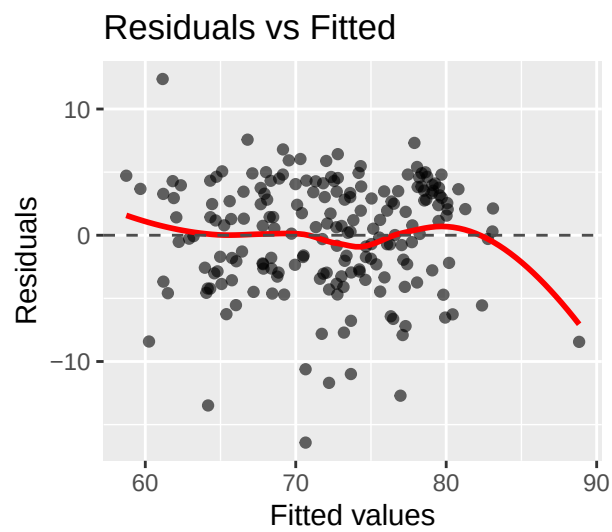
We estimate the model using heteroskedasticity robust standard errors (HC1) and report these in all subsequent specifications to account for potential heteroskedasticity.

The coefficient on log GDP per capita is positive and statistically significant ($p < 0.001$). A 10 percent increase in GDP per capita is associated with an approximately 0.48 year increase in life expectancy. This reflects a cross country relationship and should not be interpreted as causal within a given country.

4. CLM Assumptions

To evaluate the five assumptions of the classical linear model, we generate the following diagnostic plots:





1. **IID**: Country observations may not be fully independent due to shared geography, culture, and historical factors. While we do not observe clear patterns in the residuals suggesting strong dependence, this assumption is unlikely to hold perfectly in a cross country setting.
2. **Linear conditional expectation**: The residuals vs. fitted values plot shows slight systematic curvature rather than a random scatter around zero, indicating potential functional form misspecification. This suggests that GDP per capita alone does not fully capture the relationship with life expectancy.
3. **Perfect collineaerity**: The primary model includes a single explanatory variable and therefore satisfies the perfect collinearity assumption. In expanded models, we exclude average daily income due to its high correlation with GDP per capita to avoid multicollinearity.
4. **Homoskedasticity**: The Breusch-Pagan test yields a p-value of 0.505, so we fail to reject the null hypothesis of constant variance. There is no strong statistical evidence of heteroskedasticity, though we continue to use robust standard errors as a precaution.
5. **Normality of errors**: The Q-Q plot suggests that residuals are approximately normally distributed, with some deviation in the tails. This may reflect the presence of outliers or a relatively small sample size, but overall the normality assumption appears reasonably satisfied for inference.

5. Expanded Analysis

To better understand the relationship between life expectancy and broader measures of development, we extend our baseline model by incorporating additional variables capturing demographics, inequality, and population characteristics. Before constructing the expanded model, we note that several variables are strongly correlated with GDP per capita. In particular, average daily income is

highly correlated with GDP per capita (correlation ≈ 0.96), suggesting that both variables capture largely the same underlying economic variation. Including both variables would introduce multicollinearity and lead to unstable coefficient estimates. For this reason, we exclude average daily income and retain GDP per capita as our primary measure of economic development.

5.1 Demographic Factors

We extend the model by adding demographic variables for fertility (children per woman) and median age. Higher fertility is associated with lower life expectancy, while higher median age is associated with higher life expectancy.

$$\begin{aligned} \text{lex} = 46.327 &+ 2.415 \cdot \log(\text{GDP per capita}) \\ &- 1.274 \cdot \text{children per woman} \\ &+ 0.205 \cdot \text{median age} \end{aligned}$$

We observe that median age is highly correlated with life expectancy at birth (0.79) and also exhibits multicollinearity with other explanatory variables. A coefficient test shows that all three variables are statistically significant at the 0.05 level when using robust standard errors. Including these variables improves model fit and reduces the magnitude of the GDP per capita coefficient, suggesting that part of the relationship between income and life expectancy operates through demographic factors. An ANOVA test confirms that adding the demographic variables significantly improves model fit ($F = 18.85$, $p < 0.0001$), indicating that these factors add meaningful additional explanatory power.

5.2 Inequality

We next incorporate inequality by adding the Gini index, a measure of income inequality. The estimated coefficient on the Gini index is negative and statistically significant, indicating that higher inequality is associated with lower life expectancy, holding other factors constant. In this specification, median age becomes statistically insignificant ($p = 0.07$) and exhibits moderate multicollinearity with other explanatory variables (VIF $= 5.4$). We therefore exclude it from the final model, while the effects of GDP per capita and fertility remain statistically significant and consistent in direction.

$$\begin{aligned} \text{lex} = 59.131 &+ 2.660 \cdot \log(\text{GDP per capita}) \\ &- 2.061 \cdot \text{children per woman} \\ &- 0.179 \cdot \text{Gini} \end{aligned}$$

An ANOVA test comparing models with and without the inequality measure shows that adding the Gini index significantly improves model fit ($F = 15.15$, $p = 0.0001$). We conclude that inequality contributes meaningful explanatory power for life expectancy.

Conclusion

Overall, economic development is strongly associated with life expectancy, but demographics and inequality provide additional explanatory power in understanding cross country differences.

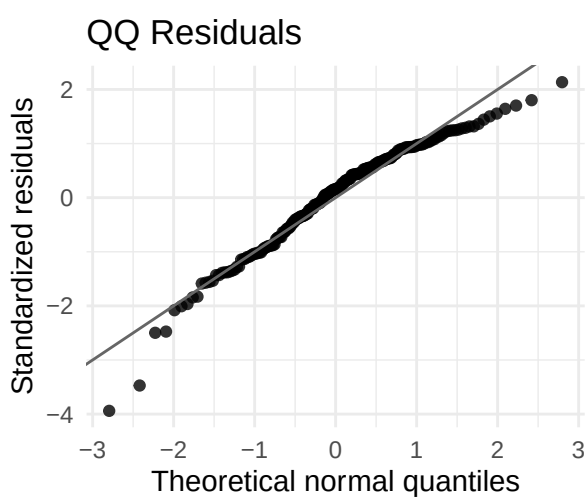
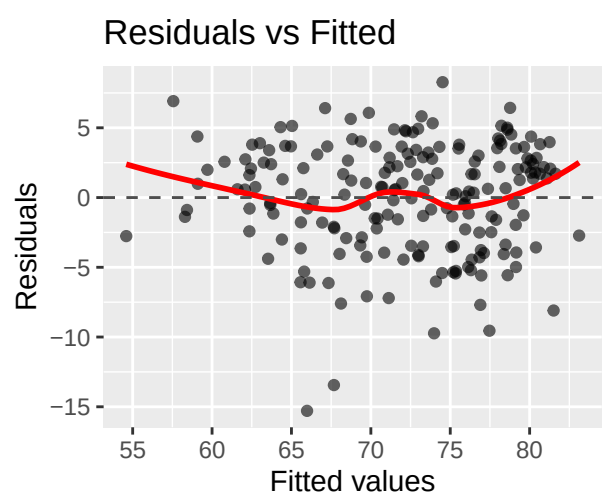
Appendix

Datasource

Our data was initially sourced from www.gapminder.org, using the interface provided on the website. After confirming our approach, we downloaded individual CSV files for each indicator from https://github.com/open-numbers/ddf–open_numbers–world_development_indicators and combined these into one dataset. The code for this wrangling process is in the data_cleaning folder in our project repo. We removed two countries without life expectancy data. **Given the overall clean dataset is small, we have included this CSV file in our repo under data/combined_data_2022.csv and recommend that the instructor access this file directly to assess the analysis presented herein.**

Model Specifications

Our question examines the relationship between GDP per capita and life expectancy. Initial visual analysis led us to apply a logarithmic transformation to GDP per capita. When considering an expanded version of the model, we first did a bivariate analysis of each variable with life expectancy (scatter plot, correlation coefficient and linear regression with coefficient test). We also considered the correlation of each variable with GDP per capita to identify potential multicollinearity concerns and excluded average daily income as a result.

We constructed our model in a step-wise fashion, starting with demographic factors (fertility and median age), and then proceeding to add the gini coefficient to capture inequality. Finally, we removed the median age variable since it becomes statistically insignificant in the presence of the gini coefficient, and exhibits multicollinearity with other explanatory variables. The remaining variables are not statistically significant and are excluded from the final model to avoid overfitting.

The final residuals vs fitted values plot shows slight parabolic curvature rather than a straight line, implying the addition of an x^2 term may improve model fit. However, we find that adding a squared term for fertility does not significantly improve model fit. The curvature may rather be driven by outliers, but we do not have a strong theoretical reason to exclude them from our analysis.